

JACK DAI

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Education

Colby College, Waterville, ME, USA

Aug, 2025 – Present

Expected Graduation: May 2029

B.A. Candidate in Mathematics and Computer Science, GPA 3.7/4.0

Relevant Coursework: CS 231 Data Structures and Algorithms, CS 252 Mathematical Data Analysis and Visualization, MA 253 Linear Algebra

Technical Projects

Digital Humanities Exhibit: YMCA and Modern Physical Education in China

Jan, 2026 – Present

Digital Humanities, Archival Research, Web Publishing

- Engineered and maintained a production digital exhibit (lllgoyour.com) with Eleventy/Nunjucks templates and reusable page components for timelines, locations, key figures, and source records.
- Built a data pipeline that compiles structured CSV source/link tables into JSON assets via Node.js scripts, with header checks and evidence-status validation to keep updates auditable.
- Maintained code and data versioning in GitHub ([LLLgoyour/across-the-pacific-gymnasium](https://github.com/LLLgoyour/across-the-pacific-gymnasium)).

CS252 Mathematical Data Analysis & Visualization Projects

Jan, 2026 – Present

- Implemented linear/multiple/polynomial regression pipelines with SciPy least-squares, normal-equation, and QR-based solvers, including residual, R^2 , and MSE diagnostics.
- Implemented K-means clustering from scratch with randomized initialization, centroid updates, inertia tracking, multi-run selection, and image-segmentation utilities.
- Maintained project code in GitHub ([LLLgoyour/cs252](https://github.com/LLLgoyour/cs252)) with clear structure and version control.

CS231 Data Structures & Algorithms Projects

Sep, 2025 – Dec, 2025

- Implemented core data structures in Java, including linked lists, stacks, queues, binary search trees, and hash maps.
- Built graph-search solutions using BFS, DFS, and A* for pathfinding and maze-solving tasks.
- Maintained project code in GitHub ([LLLgoyour/cs231](https://github.com/LLLgoyour/cs231)) with clear structure and documentation.

LUI

Aug, 2021 – Present

Vue UI Framework

- Designed and built an open-source UI framework using Vue.js and TypeScript, with reusable component patterns and shared styling primitives.
- Implemented consistent theming, composable layout abstractions, and accessibility-aware defaults to improve integration quality.
- Maintained documentation and demos in GitHub ([LLLgoyour/LUI](https://github.com/LLLgoyour/LUI)) to support contributor onboarding and component verification.

Selected Achievement

USA Computing Olympiad Platinum Division Qualifier

Dec, 2023

Research and Communication

Research Assistant, Advisor: Prof. Hong Zhang, Colby College

Nov, 2025 – Present

- Conducting historical research on the role of the YMCA in the development of modern sports and physical education in China.
- Collecting and analyzing archival documents, biographical records, and institutional histories related to Chinese sports modernization.
- Preparing a research paper under faculty supervision.

China Daily, Beijing, China

Jun, 2025 – Jul, 2025

Cultural Journalism, Features Reporter, Videographer

- Produced interview-driven cultural features, including source preparation, transcript synthesis, and narrative drafting under editorial deadlines.
- Reported on live events and developed publishable stories from raw notes, recordings, and stakeholder interviews.
- Edited short-form video coverage and iterated on structure and clarity based on editorial feedback.

Skills

- Languages: Java, Python, TypeScript, JavaScript, C++, Go, HTML/CSS
- Frameworks and Tools: NumPy, SciPy, Pandas, Matplotlib, Jupyter, Vue.js, Git/GitHub, LaTeX/Typst
- Relevant Algorithm Concepts: data structures, graph search (BFS/DFS/A*), regression modeling, QR decomposition, clustering, data transformation, debugging, complexity analysis

Additional

- Languages: bilingual in English and Mandarin Chinese